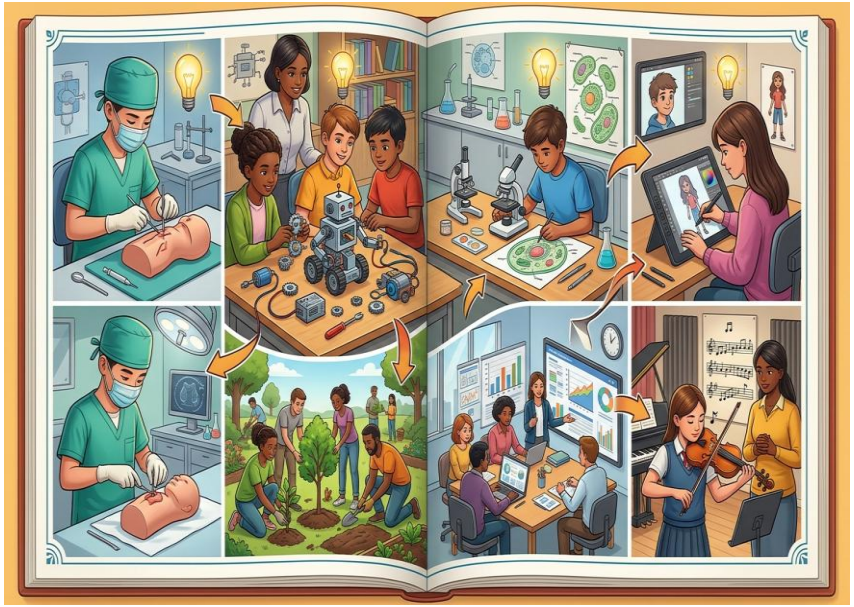


Chemistry in Action: Exploring Chemical Reactions in Our World

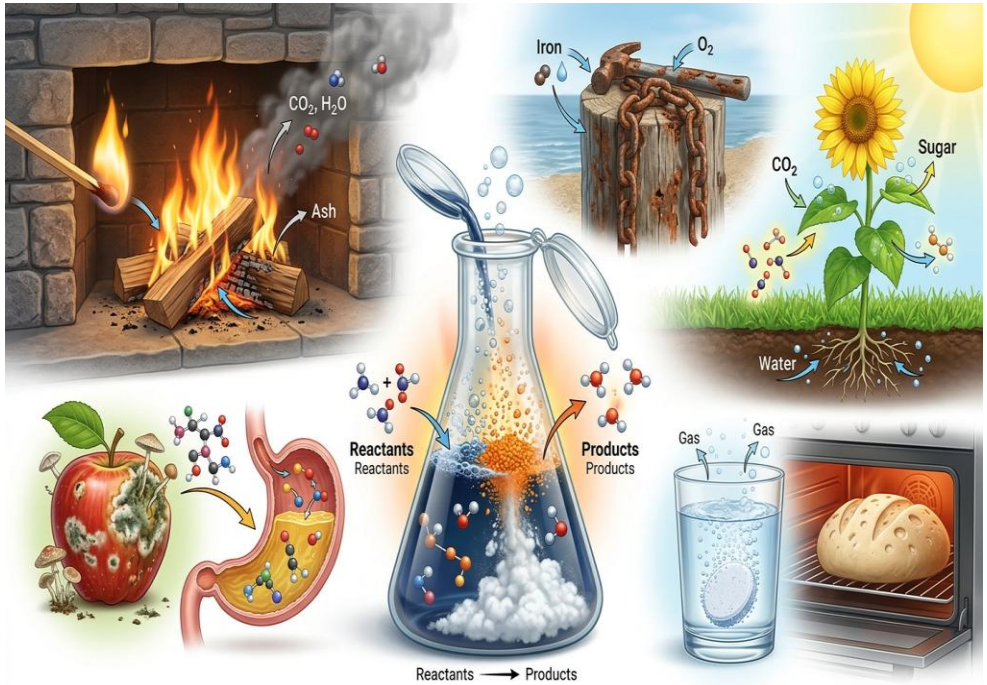
Learning Objectives

- **Explain common chemical reactions**: Decomposition, Acid on Carbonates, Acids on Metals, and Combustion.
- **Identify daily chemical reactions** in baking, cleaning, and burning.
- **Understand the chemical energy** that powers light and sound applications.



What is a Chemical Reaction?

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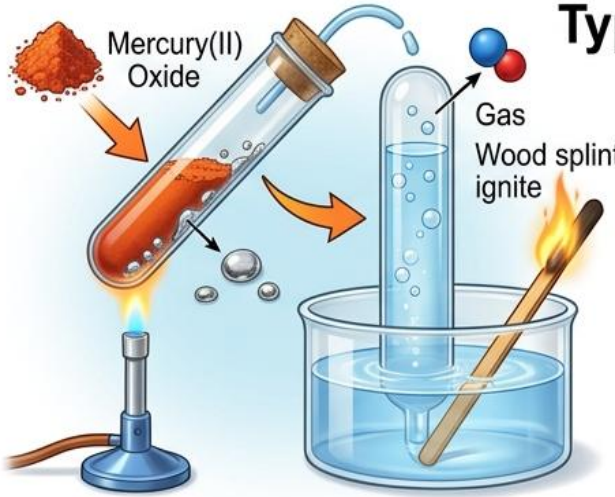


- A process where **substances** (***reactants***) transform into new **substances** (***products***).
- **Visual Example**: A glowing match.
- **Reactant**: Wood and Oxygen.
- **Product**: Ash, Smoke, and Energy.
- **Energy Output**: Light and Sound waves (linking back to LAS 10).

Type 1: Decomposition Reactions

- **Definition**: A single compound breaks down into two or more simpler substances.
- **Formula: AB → A + B**

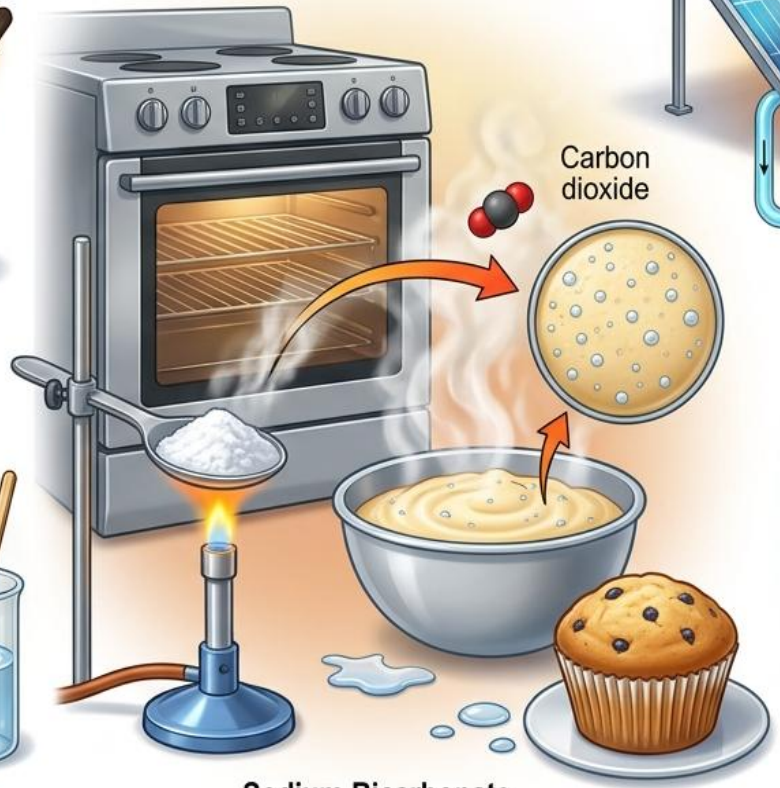
Type 1: Decomposition Reactions



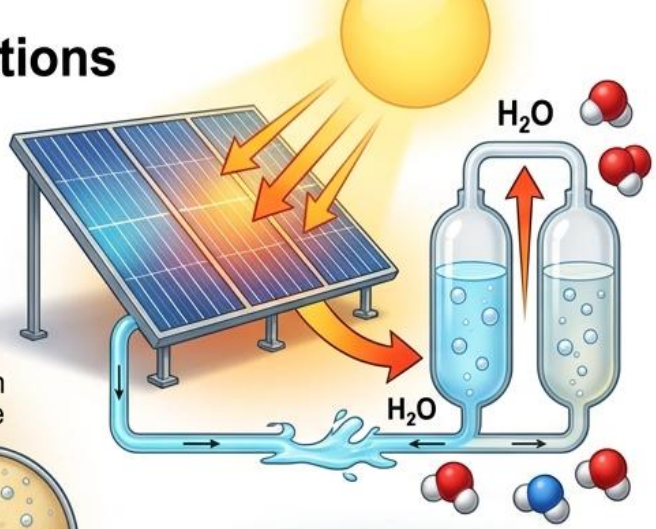
Mercury(II) Oxide (mercury) + trough)

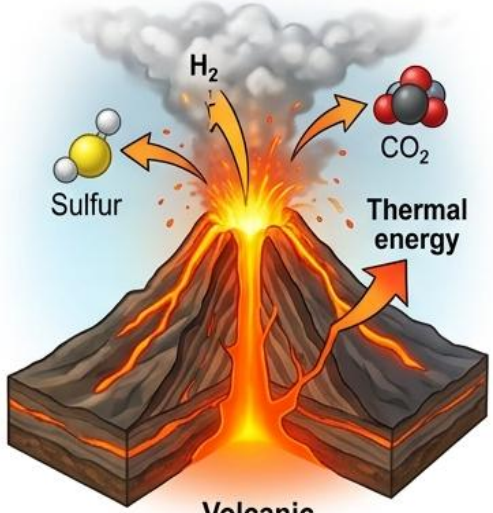


Malachite (Copper Carbonate) **Copper oxide** **Gas** (Carbon dioxide)



Sodium Bicarbonate (baking soda)





Volcanic

Daily Activity: Baking (Decomposition)



- **Reaction**: Heating Sodium Bicarbonate (Baking Soda).
- Visual Details: Imagine a tray of muffins in an oven. The heat causes the **baking soda** to **decompose**, releasing **Carbon Dioxide (CO_2) gas**.
- The CO_2 gas creates tiny bubbles, making the bread fluffy and light.

Type 2: Acid on Carbonates

- **Definition**: An acid reacts with a carbonate to produce Salt, Water, and Carbon Dioxide gas.
- **Visual Example**: Vinegar (acid) poured onto an eggshell (carbonate).
- **Visual Detail**: Immediate fizzing and bubbling as CO2 is released.



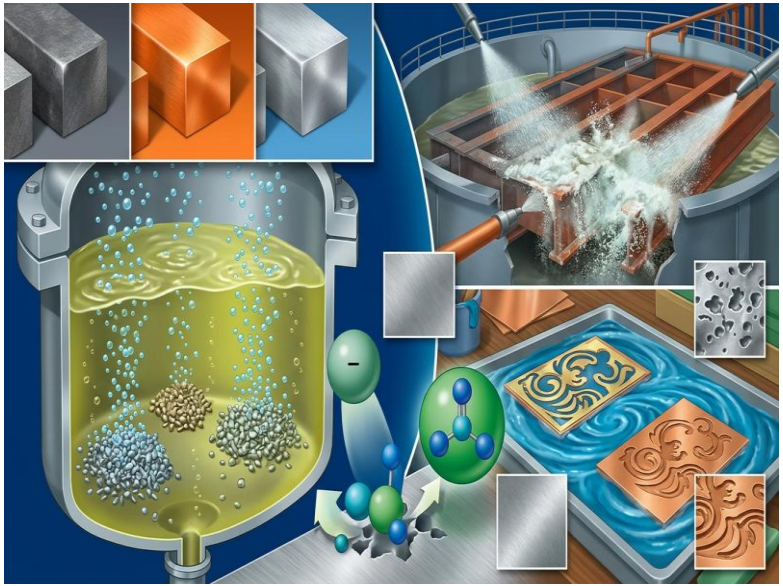
Daily Activity: Cleaning (Acid on Carbonates)



- **Application**: Removing limescale from a kettle or bathroom tiles.
- **Visual Details**: A person sprays a lemon-based cleaner (acid) on white crusty mineral deposits (carbonates).
- The **'fizz'** shows the acid is breaking down the hard mineral into soluble salts that wash away.

Type 3: Acids on Metals

- **Definition**: Acid reacts with metal to produce a Salt and Hydrogen Gas.
- **Formula**: Acid + Metal → Salt + Hydrogen Gas (H2).
- **Visual Example**: Dropping a piece of Magnesium into Hydrochloric Acid.
- **Visual Detail**: The metal seems to 'disappear' as it turns into a clear salt solution, while bubbles of hydrogen rise rapidly.



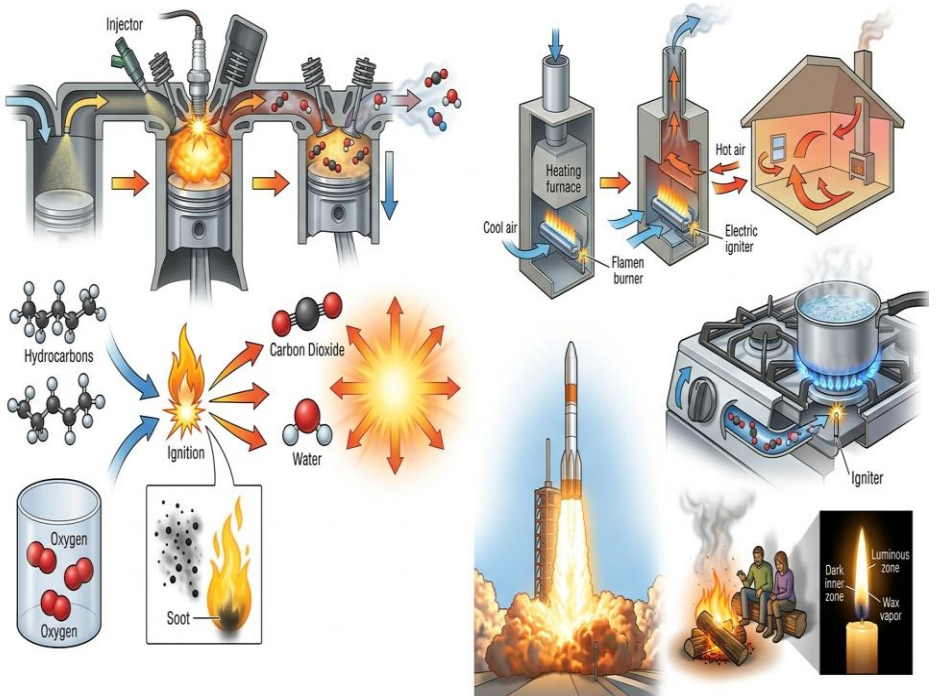
Daily Activity: Maintenance (Acids on Metals)



- **Visual Example**: Cleaning a tarnished copper penny with vinegar and salt.
- The acid reacts with the oxidized metal layer.
- **Visual Detail**: The dull, brownish penny becomes bright and shiny orange after the reaction removes the outer 'corroded' layer.

Type 4: Combustion Reactions

- **Definition**: A substance (fuel) reacts rapidly with oxygen, releasing heat and light.
- **Visual Example**: A gas stove burner.
- **Blue flame** = complete combustion.
- **Yellow flame** = incomplete combustion.

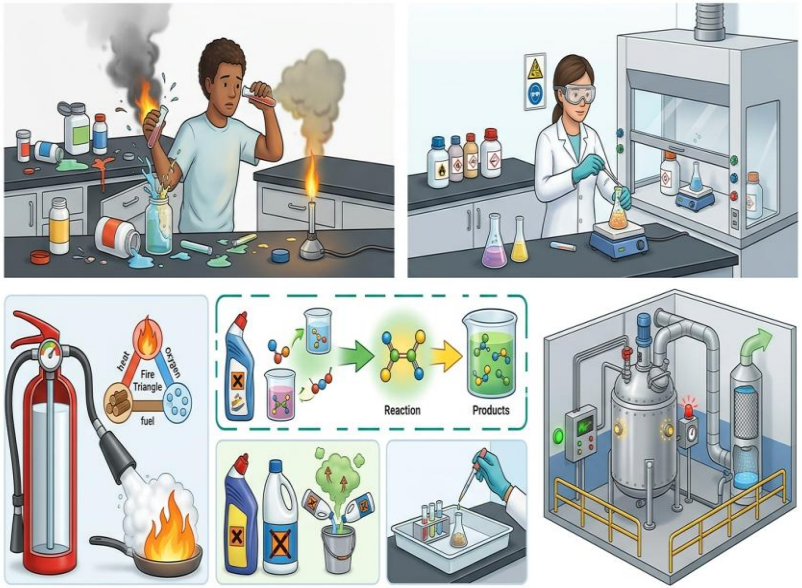


Daily Activity: Burning (Combustion)



- **Visual Example**: A candle burning in a dark room.
- **Chemical Principle**: The wax (fuel) reacts with oxygen to produce water vapor, CO_2, and energy.
- The light waves produced allow us to see (navigation/safety), just like the traffic signals in LAS 11.

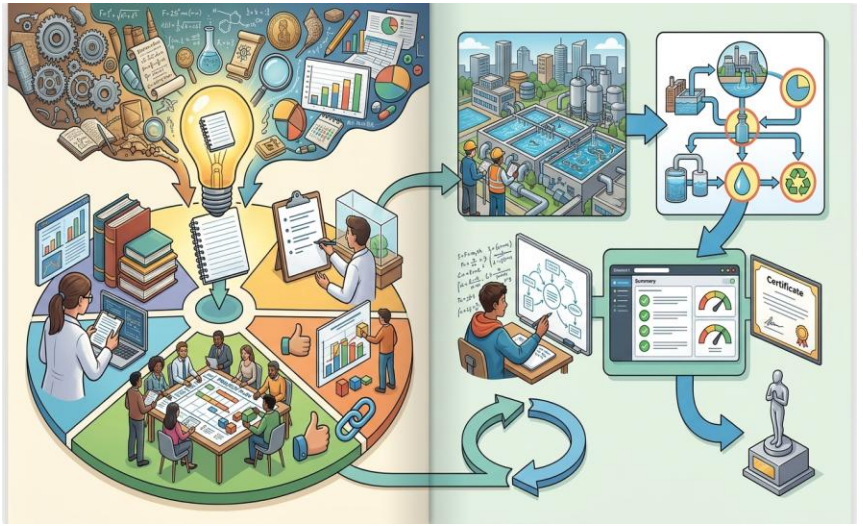
Safety and Chemical Reactions



- Combustion provides light for safety at night.
- Cleaning reactions remove pathogens for health.
- Understanding Acids prevents accidental damage to tech devices.

Summary & Review

- **Decomposition**: Breaking down (Baking).
- **Acid + Carbonate**: Fizzing (Cleaning).
- **Acid + Metal**: Gas & Salts (Maintenance).
- **Combustion**: Heat & Light (Burning).
- Chemistry is the foundation of the energy that powers our world!



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About his life and contributions